Special instructions for invigilators

Special instructions for students

Constants and Formulae

permittivity of free space:	$\varepsilon_o = 8.85 \times 10^{-12}$ F/m
permeability of free space:	$\mu_o = 4\pi \times 10^{-7}$ H/m
intrinsic carrier concentration in Si:	$n_i = 1.45 \times 10^{10}$ cm^{-3} at $T = 300$K
dielectric constant of Si:	$\varepsilon_{Si} = 11$
dielectric constant of SiO_2:	$\varepsilon_{ox} = 4$
thermal voltage:	$V_T = kT/e = 0.026$V at $T = 300$K
charge of an electron:	$e = 1.6 \times 10^{-19}$ C

$$\left.\begin{aligned} J_n(x) &= e\mu_n n(x)E(x) + eD_n \frac{dn(x)}{dx} \\ J_p(x) &= e\mu_p p(x)E(x) - eD_p \frac{dp(x)}{dx} \end{aligned}\right\}$$ Drift-diffusion current equations

$$\left.\begin{aligned} J_n &= \frac{eD_n n_p}{L_n}\left(e^{\frac{eV}{kT}} - 1\right) \\ J_p &= \frac{eD_p p_n}{L_p}\left(e^{\frac{eV}{kT}} - 1\right) \end{aligned}\right\}$$ Diode diffusion currents

$$V_0 = \frac{kT}{e}\ln\left(\frac{N_A N_D}{n_i^2}\right)$$ Built-in voltage

$$c = c_0 \exp\left(\frac{eV}{kT}\right) \text{ with } \begin{cases} c = p_n \text{ or } n_p \\ c_0 \text{ bulk minority carrier concentration} \end{cases}$$ Minority carrier injection under bias V

$$\delta c = \Delta c \exp\left(\frac{-x}{L}\right) \text{ with } \begin{cases} \delta c = \delta p_n \text{ or } \delta n_p \\ \Delta c \text{ the excess carrier concentration} \\ \text{at the edge of the depletion region} \end{cases}$$ Excess carrier concentration as a function of distance

$$L = \sqrt{D\tau}$$ Diffusion length

$$D = \frac{kT}{e}\mu$$ Einstein relation

$$C_{diff} = \frac{e}{kT} I\tau$$ Diffusion capacitance

$$i(t) = \frac{Q(t)}{\tau} + \frac{dQ(t)}{dt}$$ Time variation of current and charge.

$$\tau_t = \frac{W_B^2}{2D}$$ Base transit time

$$\delta c_B = C_1 \exp\left(\frac{x}{L_B}\right) + C_2 \exp\left(\frac{-x}{L_B}\right)$$

$$C_1 = \frac{c_B(W_B) - c_{B_0} - \left(c_B(0) - c_{B_0}\right)\exp\left(\frac{-W_B}{L_B}\right)}{2\sinh\left(\frac{W_B}{L_B}\right)}$$

$$C_2 = \frac{\left(c_B(0) - c_{B_0}\right)\exp\left(\frac{-W_B}{L_B}\right) - \left(c_B(W_B) - c_{B_0}\right)}{2\sinh\left(\frac{W_B}{L_B}\right)}$$

Excess minority carrier concentration in base of BJT. EB junction at x=0 ; BC junction at x=W_B. c_{B0}: equilibrium concentration. L_B: minority carrier diffusion length in base.

$$w_n = \sqrt{\frac{2\varepsilon}{e}\frac{N_A}{N_A N_D + N_D^2}(V_0 - V)}$$

$$w_p = \sqrt{\frac{2\varepsilon}{e}\frac{N_D}{N_A N_D + N_A^2}(V_0 - V)}$$

Depletion regions in pn diode

$$W_{depl} = \sqrt{\frac{2\varepsilon}{e}\frac{N_A + N_D}{N_A N_D}(V_0 - V)}$$

The Questions

1.

a) The current in pn diodes and BJTs at low forward bias conditions is determined by minority carriers because of (choose one of the following):

i) drift of minority carriers through the neutral device regions.

ii) generation of minority carriers in the neutral device regions.

iii) diffusion of minority carriers through the neutral device regions.

iv) generation of majority carriers in the neutral device regions.

Explain your answer briefly. [5]

b) Short and long material approximations are simplifications of the correct solution of the continuity equation for minority carriers. The three solutions are plotted in Fig. 1.1 for a 300 nm long material. The mobility of the minority carriers is $\mu = 120$ cm^2 V^{-1} s^{-1} and their lifetime is $\tau = 2.88\ 10^{-10}$ s. Which of the approximations gives the smallest error in the current calculation? Explain your answer briefly. [5]

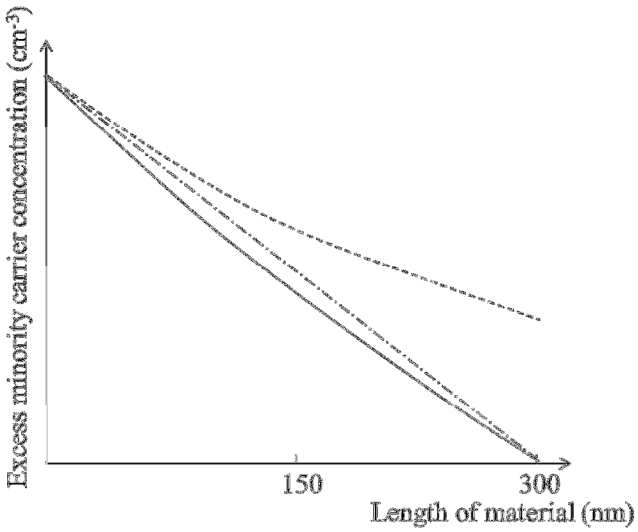


Figure 1.1 Excess minority carrier concentrations. The solid line shows the correct solution; the dashed-dotted line shows the linear approximation and the dashed line shows the exponential approximation.

c) The measurement of the time response of the pn-diode switched from forward to reverse bias is given in Fig. 1.2 (see next page). Sketch, on one figure, the excess minority carrier concentration in one layer of the pn diode as a function of distance, at times $t = t_1$, $t = 0$, $t = t_{sd}$ and $t = t_2$. Assume that the layers are long. [5]

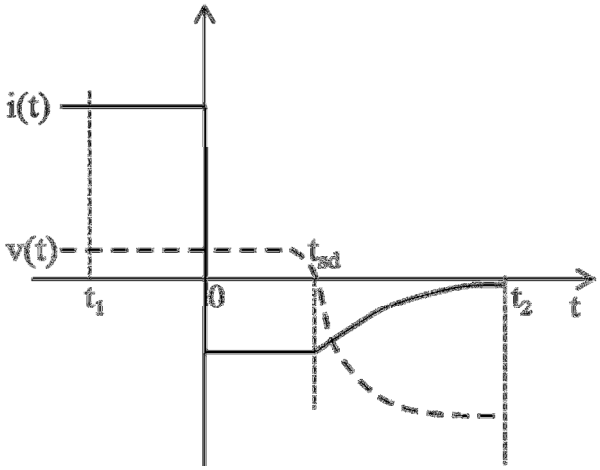


Figure 1.2 The variation of the current *i(t)* and voltage *v(t)* of a switching pn diode.

d) Fig. 1.3 gives a sketch of the minority carrier concentration in the base layer of a pnp BJT.

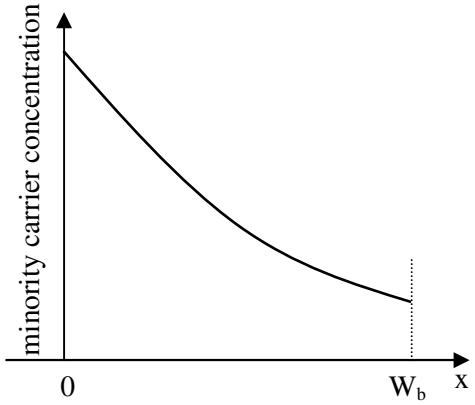


Figure 1.3 Variation of the excess minority carrier concentration in the base.

Copy Fig. 1.3 and add the correct label on the y-axis (δp_n or δn_p). Use your sketch and the definition of diffusion current to explain how the recombination current can be calculated. [5]

2.

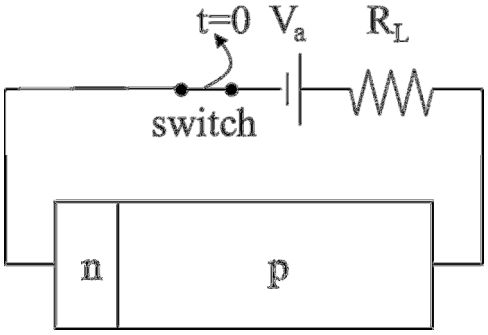


Figure 2.1 Bias circuit for a pn diode with switch.

Table 2.1: Material parameters for the diode in Fig. 2.1.

Material parameters	n-region	p-region
Doping density (cm^{-3})	8×10^{19}	2×10^{17}
Region length (μm)	0.25	500
Minority carrier lifetime (s)	9×10^{-10}	1×10^{-6}
Minority carrier diffusion length (cm)	3×10^{-5}	5×10^{-3}
Cross sectional area (μm^2)	100	100

Temperature is T = 300K.

a) Calculate the current flowing through the diode of Fig. 2.1 when a forward bias voltage of V_d = 0.5 V appears across the diode. [5]

b) At t = 0, the switch in the circuit of Fig. 2.1 is opened. Derive the expression of the time variation of the voltage across the diode, $v_d(t)$. Define all parameters and explain all assumptions you make. [7]

Hint: sketch the excess minority carrier concentration variation. The excess charge concentration variation is : $Q(t) = I\,\tau\,exp(-t/\tau)$.

c) Calculate the time needed for the voltage across the diode to settle to 0. [3]

3.

a) Explain briefly the meaning of the term bipolar in the name of the device: bipolar junction transistor (BJT). [3]

b) Assume that an npn BJT has an emitter efficiency of $\gamma = 1$ and that the base is short though recombination is taking place. If the excess minority carrier charge in the base is given by Q_B, give the expression for the base current I_B and the collector current I_C as a function of Q_B. Define all parameters that you use. [5]

c) Calculate the current gain, β in the npn BJT in forward active mode with $\gamma = 1$. The material parameters are listed in Table 3.1. Take the influence that the reverse biased BC junction has into account. The applied voltages are V_{EB} = 0.5 V and V_{BC} = -1V. The temperature is T = 300 K. [7]

Table 3.1 Material parameters of the bipolar junction transistor.

Material parameters	emitter	base	collector
Doping density (cm^{-3})	10^{20}	10^{17}	5×10^{16}
Region length (μm)	0.1	1	500
Diffusion constant ($cm^2\ s^{-1}$)	2	6	20
Minority carrier lifetime (s)	1×10^{-10}	1×10^{-6}	1×10^{-2}
Cross sectional area (μm^2)	100	100	100

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